

Chat Filter Accuracy Audit

The filter searches for each banned word **anywhere in a message, including inside longer words**, and the word list is **mostly Brazilian Portuguese**. Its two-letter entries are applied to all eight languages the game ships in, so the filter blocks ordinary words in every one of them.

thank

DELIVERED

than^{ks}

REFUSED

do^{cu}ment

REFUSED

se^{cu}rity

REFUSED

cu is a rude word in Portuguese. ks is a two-character entry. The filter has no rule about word boundaries, so it blocks any message containing a word that holds those letters.

15,307

MESSAGES SENT

343

WORDS CONFIRMED
CENSORED

8,414

ENGLISH WORDS AFFECTED

16

RULES IDENTIFIED

Content notice. Establishing what the filter holds requires naming the terms it matches, so this report quotes profanity and slurs. They appear as evidence, not as commentary.

Black-box measurement against the live service, 11–12 September 2026. No client modification, no packet injection, no server or source access — the audit observes exactly what a player observes. All 15,307 probe records are published with this report.

The filter matches inside longer words

The filter blocks a message wherever the letters of a banned word appear. We confirmed this by placing the letters inside a made-up word, so that a block could only be caused by those letters and not by the ordinary word that first revealed the problem.

WE SENT THIS	WHAT HAPPENED	WE SENT THIS	WHAT HAPPENED	WHAT THIS SHOWS
cu	blocked	ks	blocked	these letters are on the word list
xxcuxx	blocked	xxksxx	blocked	the filter blocks them inside another word
xxcxx	delivered	xxkxx	delivered	the first letter alone is allowed
xxuxx	delivered	xxsxx	delivered	the second letter alone is allowed
xxucxx	delivered	xxskxx	delivered	the same letters reversed are allowed

The filter delivered 10,627 words during this audit. **None of those 10,627 words contains cu or ks.** Every word we tested that contains those letters was blocked.

Short entries affect far more words

Ordinary English words blocked, by entry. A short entry appears by chance inside many more words than a long one does.

cu		5,016	document, discuss, security
nb		956	number, inbox, unbind
meter		695	parameter, diameter
anta		419	advantage, santa, fantastic
ks		408	thanks, tasks, books, works
pus		230	campus, octopus, push
pica		227	typical, tropical
puta		95	reputation, computation

Eight further entries account for the remainder.  two-character entry  longer entry

Coverage is uneven by language, not uniformly strict

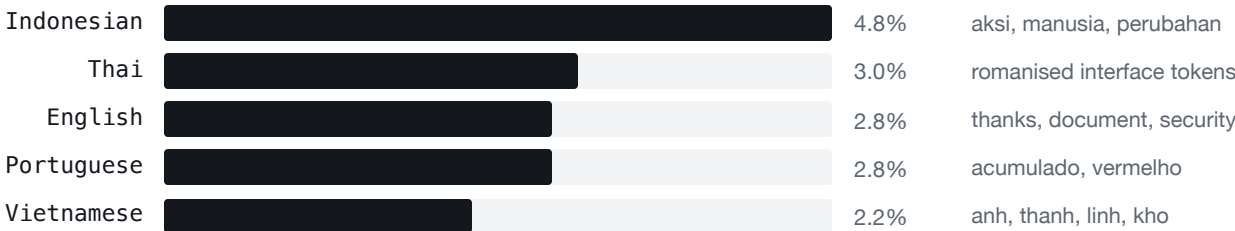
Twenty-seven Portuguese profanity terms were tested and **all twenty-seven were blocked**, down to low-frequency items like arrombado and otario. Tested identically:

bitch DELIVERED	rape DELIVERED	slut DELIVERED	whore DELIVERED
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Both halves share one cause: a term list that has not been reviewed against the locales it is applied to.

Every market, and the game's own interface text

The block rate was measured separately against the common vocabulary of each shipped language. It is uniform — this is not an English-only problem.



The service censors words from its own shipped localisation files. A player who types a word they can read on screen is told their speech is inappropriate.

What the player experiences

The message is not delivered and the sender sees a modal stating the speech is inappropriate. Nothing indicates which word caused it, and the offending sequence is not visible as a word — so the rule cannot be learned and cannot be avoided. Repeated blocking of ordinary words read as the game being broken rather than as moderation.

Why this costs support time

PROPERTY	CONSEQUENCE
Affects high-frequency vocabulary	thanks is among the most common words in social game chat, and the filter blocks it
Silent and unattributable	Players cannot infer the rule, so behaviour does not adapt
Not individually actionable	Each report names a different ordinary word produced by the same handful of rules
Asymmetric by market	Indonesian shows the highest rate of the five vocabularies measured

Confidence in these figures

FIGURE	BASIS
343 words confirmed censored	Measured. Each required two separate tests showing a block against a measured 1.2% per-probe error rate.
8,414 English words affected	Derived. Confirmed rules projected across a dictionary; these specific words were not each sent.
16 rules	A lower bound. Entries whose letters did not occur in the 11,868 strings tested would not have surfaced.

Four options. None of them blocks less.

Every option below removes false positives only — no entry is taken off the block list. They are independent and can ship in any order.

APPROACH	TOUCHES	RESIDUAL RISK	EFFORT
A Ship the confirmed exception list	Data only	Words outside the tested set	Hours
B Add the rescue shim	One call site	Names and slang not in a dictionary	1 day
C Minimum-length boundary rule	Matcher condition	Long entries still match inside longer words	1–2 days
D Replace the matcher	Component swap	None of this class	1–2 weeks

A · Exception list — no code change

If the filter already consults an exception list, this is a data deployment. `words-to-allow.txt` holds the 343 confirmed words, one per line, each annotated with the rule responsible.

C · Boundary rule — best value per line

One condition addresses the cause rather than the symptoms. Long entries such as `caralho` behave exactly as they do today, so strength against substantial terms is untouched.

Entries shorter than five characters match only as whole words. Entries of five characters or more are unchanged.

With this rule in place, the filter still blocks a message that is only the word `cu`, and it stops blocking `document`. The filter still blocks a message that is only the word `ks`, and it stops blocking `thanks`.

Recommended rollout

- 1 Shadow.** Evaluate the new decision alongside the current one and log where they differ, without changing what players see.
- 2 Check the direction of every difference.** All differences must be messages the filter blocks today that would now be delivered. A message newly *blocked* indicates a wiring error, not a tuning question — none of these options adds blocking.
- 3 Enable.** A and B may ship independently and in either order.

Three lines at one call site

Call the shim where the existing filter has decided to block and knows the matched span. If that span sits inside ordinary vocabulary, suppress the block. Your matching logic, word list, severity rules and protocol are all unchanged.

```
# once, at startup
from rescue import Rescue
rescue = Rescue("allowlist-en.txt")

# inside your filter, where you were about to block:
if rescue.is_rescued(text, match_start, match_end):
    pass          # ordinary word – deliver the message
else:
    block()       # unchanged behaviour
```

Cost: 473,532 entries as a hash set, loaded once. Lookup is a single hash probe — not measurable against a network round trip.
Why it works without knowing your list: you assert what is legitimate rather than identifying which entries are at fault, so it holds however the list changes.

The boundary rule, language-agnostic

```
// term matched at [start, end) in the message
func accept(term, message, start, end) {
    if (len(term) >= MIN_SUBSTRING_LEN) return true;    // 5 – unchanged

    // short term: require a word boundary on both sides
    before = (start == 0)          ? null : message[start-1];
    after  = (end == len(message)) ? null : message[end];
    return !isLetter(before) && !isLetter(after);
}
```

If your matcher is a compiled automaton, tag short patterns at build time and apply the boundary test only to those, so throughput is unaffected for the rest.

What ships in this package

PATH	FORMAT	SIZE	USE
findings/words-to-allow.txt	one word per line	343	Option A payload
findings/affected-words.txt	one word per line, by rule	8,414	Option A, broader set
fix/allowlist-en.txt	one word per line	473,532	Option B corpus
fix/rescue.py	Python 3, no dependencies	138 In	Option B shim
engine/chatguard.py	Python 3, no dependencies	676 In	Option D matcher
engine/vectors/golden.jsonl	JSONL	29	Conformance for any port

Confirm the defect in ten seconds

Send thank in game, then thanks. The first is delivered, and it blocks the second. This needs nothing from this package.

Must become deliverable

INPUT	BLOCKED BY
thanks	ks
document	cu
number	nb
advantage	anta
parameter	meter
reputation	puta

Must stay blocked

INPUT	NOTE
cu	the entry itself, as a word
ks	the entry itself, as a word
puta	short entry, still matches
caralho	long entry, rule does not apply
arrombado	long entry, unaffected

If any of these becomes deliverable the change was wired incorrectly — no option removes an entry from the block list.

```
python3 tests/test_all.py      # 55 tests
python3 tools/conformance.py   # 29 language-agnostic vectors
python3 tools/check_numbers.py # every figure here traces to the evidence
```

06 · METHOD & LIMITATIONS

Messages were sent through the retail client into a private party channel, one at a time, and read back from chat history. Three controls govern what reached this report:

CONTROL	WHY
We report a group of letters only after testing it inside a made-up word	The filter blocks cucumber, but that does not show which part caused it
Two separate tests must both show a block	Measured per-probe error rate is 1.2% (6 spurious in 491), so one reading is not a result
A message that was never sent counts as untested, not delivered	All but 15 were retried to a real verdict; a false-negative audit over 250 words returned zero misses

LIMITATION	CONSEQUENCE
Rule set is a lower bound	There may be further entries the tested vocabulary did not reveal
Dictionary counts are derived	8,414 is a projection; the 343 measured words are kept separately
Point-in-time	Describes the service on 11–12 September 2026; a list edit changes it
Non-Latin locales under-covered	Korean, Chinese and Japanese were not swept; no conclusion is offered

Every quantity here derives from `findings/findings.json` and is checked by `tools/check_numbers.py`, which fails if a published figure is not supported by the evidence. All 15,307 probe records: `findings/raw-logs/`.